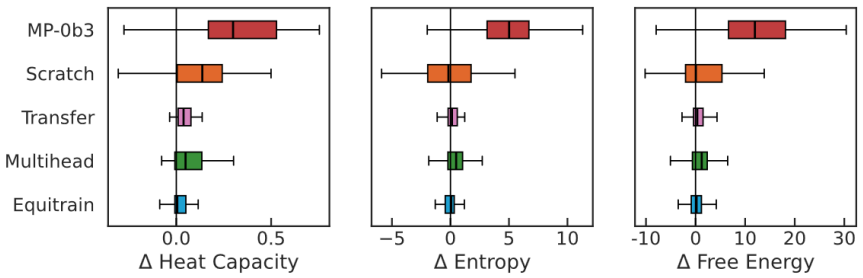
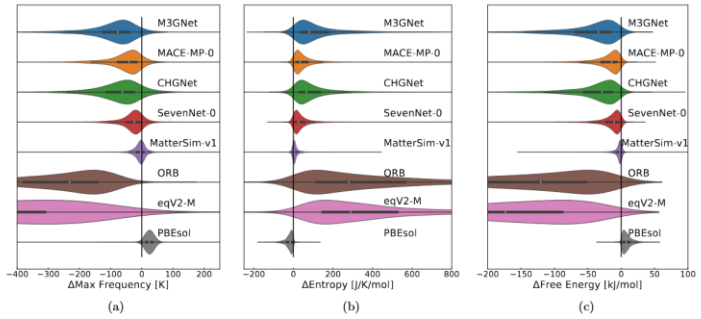
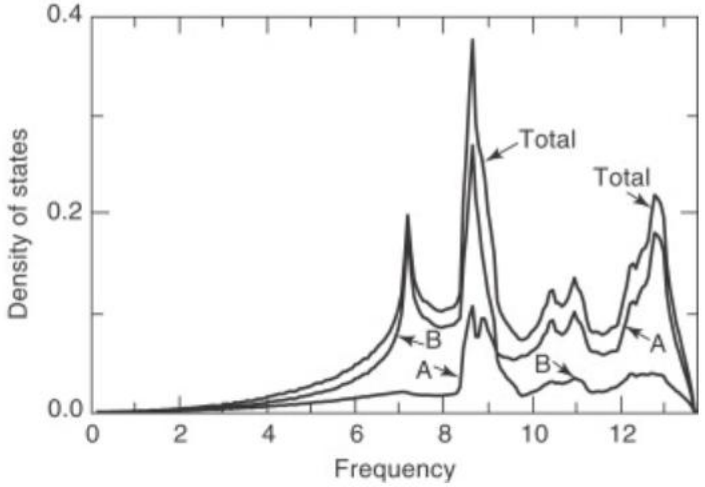
Project Overview:

- Based on Grandel, Benner & George, BAM Berlin, March 2026
- DFT -> Finetuning -> MD/Thermal Properties
- **DATA**: Generate rattled structures, relax w/ MLIP, DFT along the relaxation trajectories
- **FINETUNE**: Use MatterTune to finetune models like MACE or Mattersim
- **Analyze**: Use LAMMPS for MD and Phonopy with MLIP as the calculator

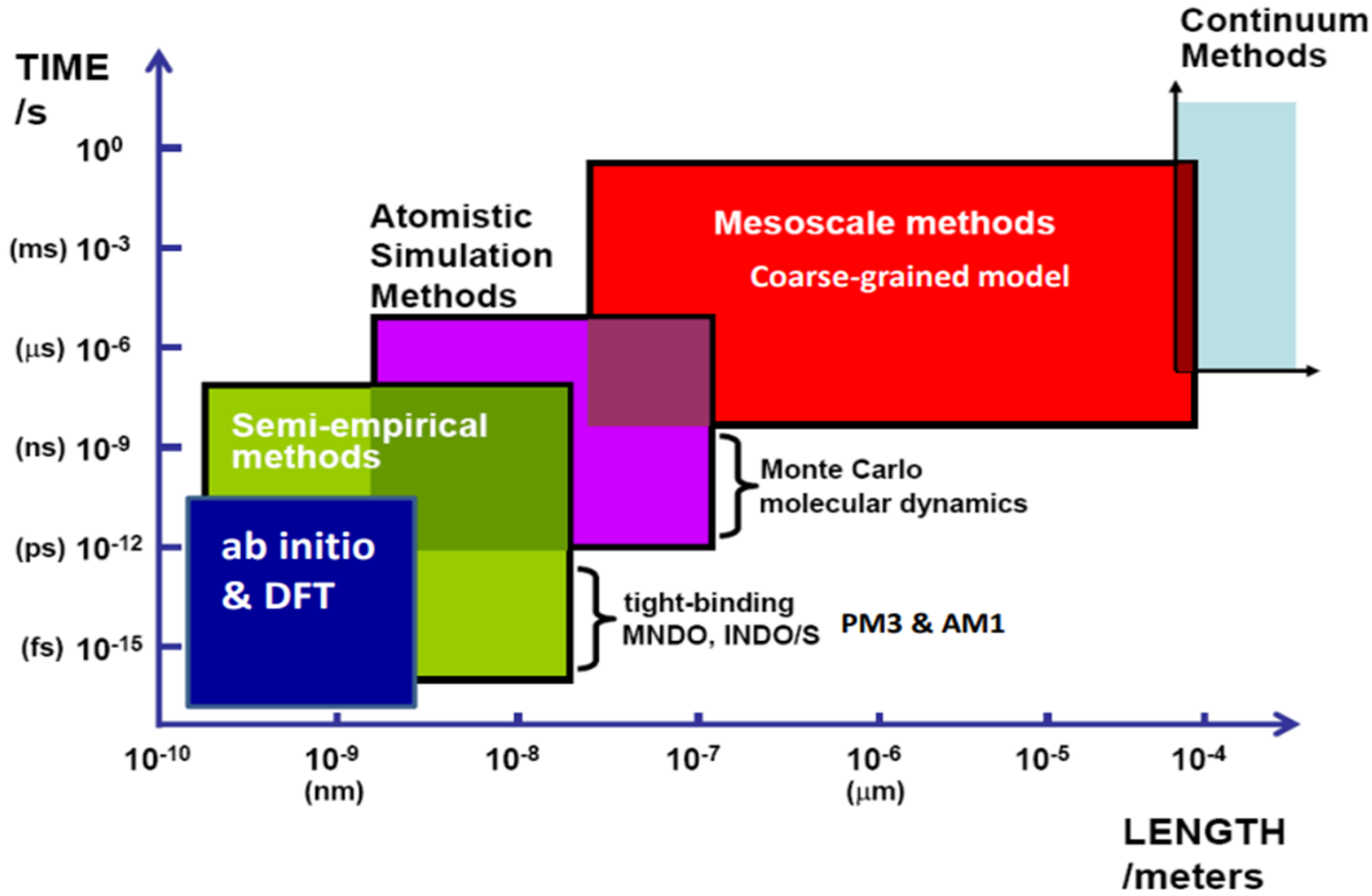
- Materials: Si, GaAs (abundant thermos data)







Hierarchical Multi-physics & Multi-scale Modeling Paradigm



Density Functional Theory (DFT):

- Ab-initio vs Semi-empirical
- Schrodinger's Equation: $\hat{H}\Psi = E\Psi$ (solve for wave function)
- Can only solve Schrodinger Equation for 2 bodies (H atom)

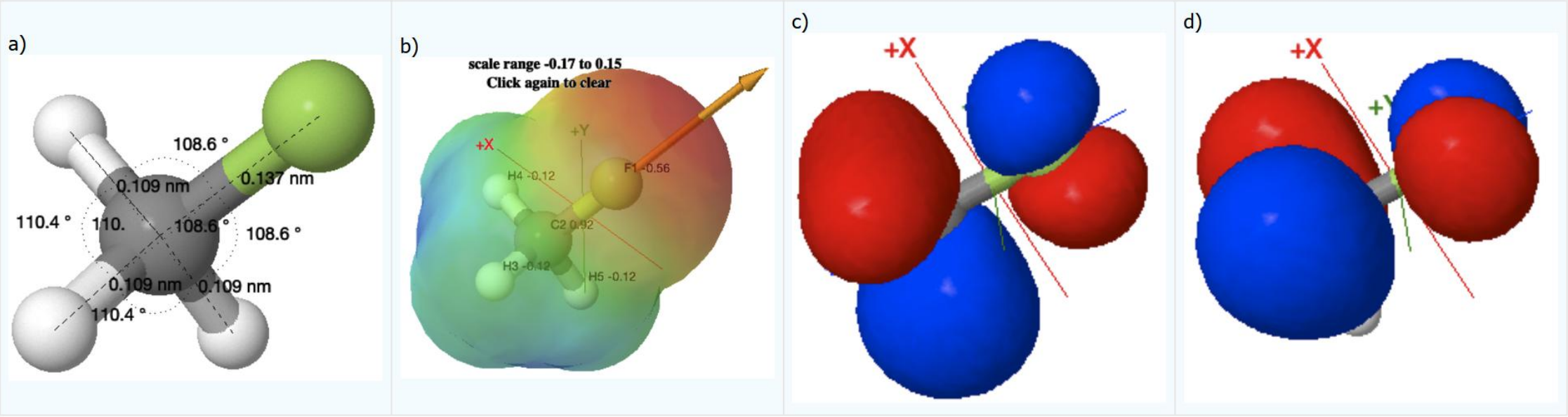
- Hohenberg-Kohn theorem – Electron density map contains all ground state information (0 K in Kohn-Sham DFT)
- Exchange-correlation functional: $\mathbf{E_{xc} = E_x + E_c + \Delta T}$
  - Contains all messy 3N physics, needs to be approximated
- DFT calculations run in $O(N^3)$ (Not as bad as CCSD(T) at $O(N^7)$)!!

- Example Functionals: PBE, B3LYP, BLYP are used to improve speed/accuracy
- Built around basis sets which are used to represent orbitals in wave function

Geometry

Dipole Moment

HOMO/LUMO





PACKMOL







Gaussian-Type Orbital Basis Sets



Plane Wave Basis Sets

Quantum Espresso VM:

- Virtual Machine - Computer in your computer (mem alloc, etc.)
- Don’t need a GUI – interact through terminal (lighter)

- Runs on Ubuntu (Linux)
  - Modular and allows for no GUI

- Terminal (Win Term, WezTerm, macOS Term): text in -> out
- Shell – underlying program that interprets commands (bash/zsh)
- CLI – what you type commands into

Terminal

```
pwd
ls
ls -la
cd target
cd ..
cd ~
```

```
/
├── bin/
├── etc/
├── home/
│   ├── alice/
│   └── bob/
├── usr/
└── tmp/
```

```
- rwx r-x r--  1 alice physics  1024 Sep 29 18:30 run_qe.sh
|  |   |   |
|  |   |   └── Others (World) Permissions: Read-only (r--)
|  |   └────── Group Permissions: Read & Execute (r-x)
|  └────────── Owner Permissions: Read, Write & Execute (rwx)
└───────────── File Type (- = regular file, d = directory)
```

```
echo $PATH
```

```
mkdir my_project    # C
touch notes.txt     # C
cp notes.txt copy.txt
mv notes.txt draft.txt
```

```
rm draft.txt
rm -rf my_project
```

SSH!!

Text editors

Miniconda/ Package Managers
(Dependency Hell!)

Git and Github:



v.s.



Working Dir -> Staging -> Git Repo -> Github remote

```
# 1. Initialize a new local repository inside a project folder
git init

# 2. Check current status (which files are untracked or modified)
git status

# 3. Stage files for the next commit
git add parse_output.py      # Stage a specific file
git add .                    # Stage all modified/new files in the directory

# 4. Save staged changes to history with a descriptive log message
git commit -m "Add output energy parser for Quantum ESPRESSO SCF runs"

# 5. View history of commits
git log --oneline
```

```
git remote add origin https://github.com/USERNAME/qe_project.git
git branch -M main
git push -u origin main
```

```
git pull origin main
```

Quantum Espresso:

- Expand Memory and Processors
- Set up ssh with sudo apt install openssh-server –y and sudo systemctl enable --now ssh
  - .ssh/config
  - max and moritz
  - scp -P 2222 example.in max@127.0.0.1:/home/max/projects/
  - scp -P 2222 max@127.0.0.1:/home/max/projects/comp_mat/example.out .
- Set up shared folder, sudo usermod -aG vboxsf max
- Download PBEsol SSSP
  - tar -xzf SSSP-lib-pbesol-eff-v2.tar.gz
- Run basic dft with file on github
  - mpirun -np 4 pw.x -in si.scf.in > si.scf.out
  - mpirun -np 4 --mca btl_tcp_if_include lo pw.x -in si.scf.in > si.scf.out